

NFC BIO Branch Successor Pilot

v29 — L2 Generated Shell

Generated governance shell

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BIO Branch Successor Pilot

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Governance notice. This is a machine-generated L2 branch-pilot shell. It is *not* a supersession of the canonical NFC Biology Branch Book. Full supersession requires L5 gate passage per the NFC supersession policy.

BIO Branch Successor Pilot

This generated packet is an L2 branch-pilot shell for the Biology branch. It is not a supersession of the canonical BIO Branch Book. The pilot preserves the nondegenerate $C_{\text{Bio}}^{\text{nd}}$ endpoint scope without claiming evolutionary closure or full-class biology certification. The no-smuggling boundary is maintained: all structure derives from role carriers and source descendants; no biological primitives are imported as assumptions.

Pilot posture and boundary conditions

Computed BIO posture: **conditional-cert-close**. The endpoint is scoped to the nondegenerate certified biological class $C_{\text{Bio}}^{\text{nd}}$. The boundary obligation **ob:BIO.BND-open** (BND for the full replicator class) remains active and is intentionally preserved in this pilot. Evolutionary dynamics (ob:BIO.EVO) and multicellular organization (ob:BIO.MULTI) are downstream of the endpoint and are not claimed here.

No-smuggling invariant

The following primitives are *not* imported in this pilot: genes, cells, fitness functions, thermodynamic entropy, membrane biophysics, Lipschitz geometry, or any biological vocabulary not derivable from role carriers and NFC source-descendant structure.

Table 1: BIO successor-pilot claim catalog

ID	Kind	Declared	Computed	Scope
ob:BIO.REP	obligation	C	C	branch
ob:BIO.MET	obligation	C	C	branch
ob:BIO.FID	obligation	C	C	branch
ob:BIO.NC	obligation	C	C	branch
ob:BIO.HER	obligation	C	C	branch
ob:BIO.BND	obligation	C	C	scoped-nondegenerate

ID	Kind	Declared	Computed	Scope
thm:BIO.REP	theorem	C	C	branch
thm:BIO.MET	theorem	C	C	branch
thm:BIO.FID	theorem	C	C	branch
thm:BIO.NC	theorem	C	C	branch
thm:BIO.HER	theorem	C	C	branch
thm:BIO.BND	theorem	C	C	scoped- nondegenerate
thm:BIO.END	theorem	C	C	branch
thm:BIO.MULTI.twocell	theorem	C	C	branch
ob:BIO.BND-open	frontier	O	O	open-class
status:BIO	status_note	conditional- cert-close	conditional- cert-close	branch

Generated claim shells

Obligation: Replication Certification

Machine ID. ob:BIO.REP

Declared status. C **Computed status.** C

Scope. regime=nondegenerate-bio; domain=BIO-branch; certificate_scope=branch; promotion_ceiling=C

Show that the replication kernel Φ_{Bio} is sourced through U and satisfies the carrier-descent conditions. The kernel must be derived from the SCC role-carrier machinery without importing genes, cells, or primitive biological vocabulary.

Dependencies. thm:SCC.MCS

Obligation: Metabolic Bookkeeping

Machine ID. ob:BIO.MET

Declared status. C **Computed status.** C

Scope. regime=nondegenerate-bio; domain=BIO-branch; certificate_scope=branch; promotion_ceiling=C

Show that the metabolic transport satisfies additivity: the metabolic ledger W_{met} is an additive accounting of energy-equivalent transport across the certified replication boundary. No thermodynamic primitives may be imported outside the NFC carrier framework.

Dependencies. ob:BIO.REP

Obligation: Replication Fidelity Margin

Machine ID. ob:BIO.FID

Declared status. C **Computed status.** C

Scope. regime=nondegenerate-bio; domain=BIO-branch; certificate_scope=branch; promotion_ceiling=C

Show that the fidelity margin δ_{fid} separates the replicating class from the non-replicating class with a positive gap on the nondegenerate regime $C_{\text{Bio}}^{\text{nd}}$.

Dependencies. ob:BIO.REP

Obligation: No-Hidden Hereditary Channel

Machine ID. ob:BIO.NC

Declared status. C **Computed status.** C

Scope. regime=nondegenerate-bio; domain=BIO-branch; certificate_scope=branch; promotion_ceiling=C

Show that there is no undeclared hereditary channel: every hereditary pathway is visible in the declared replication kernel $\backslash\Phi\{\}_\text{Bio}$. No hidden pathway imports biological structure outside the carrier record.

Dependencies. ob: BIO.REP

Obligation: Hereditary Descent

Machine ID. ob: BIO.HER

Declared status. C **Computed status.** C

Scope. regime=nondegenerate-bio; domain=BIO-branch; certificate_scope=branch; promotion_ceiling=C

Show that the hereditary transport H_Bio passes the Book V branch-legitimacy test: descent from U through the declared replication functor, with no hidden supplementation.

Dependencies. ob: BIO.REP, ob: BIO.NC

Obligation: Boundary Self-Organization (Nondegenerate Regime)

Machine ID. ob: BIO.BND

Declared status. C **Computed status.** C

Scope. regime=nondegenerate-bio; domain=BIO-branch; certificate_scope=scoped-nondegenerate; promotion_ceiling=C

Show that interface self-organization is sourced from the Book II capacity law on the nondegenerate regime: the boundary structure B_Bio derives from bounded observational channel capacity without importing Lipschitz geometry, membrane biophysics, or cell-biology primitives as assumptions.

Dependencies. thm: BIO.REP, thm: VII.6.4

Theorem-like claim: Replication Kernel Certification

Machine ID. thm: BIO.REP

Declared status. C **Computed status.** C

Scope. regime=nondegenerate-bio; domain=BIO-branch; certificate_scope=branch; promotion_ceiling=C; constraints=SCC-role-carrier-machinery

The replication kernel $\backslash\Phi\{\}_\text{Bio}$ is a lawfully source-descended replication operator on the nondegenerate biological class C_Bio^{nd} , satisfying: (i) $\backslash\Phi\{\}_\text{Bio}$ factors through the universal source U via the declared branch realization functor; (ii) the kernel's carrier structure descends from the SCC role-carrier machinery without adding undeclared biological primitives; (iii) the replication image is observable under the declared probe family.

Dependencies. thm: SCC.MCS, thm: VII.6.4

Discharge edges. ob: BIO.REP (full)

Proof stub. Apply SCC minimal carrier structure (thm: SCC.MCS) to derive the role-carrier basis. The replication carrier is constructed as the minimal carrier satisfying the declared replication closure conditions. Nondegeneracy of C_Bio^{nd} guarantees the positive separation needed for carrier well-definition. No biological primitives are introduced: replication is defined in terms of carrier-descent and probe-response equivalence, not in terms of genetic or cellular vocabulary.

Theorem-like claim: Metabolic Ledger Theorem

Machine ID. thm: BIO.MET

Declared status. C **Computed status.** C

Scope. regime=nondegenerate-bio; domain=BIO-branch; certificate_scope=branch; promotion_ceiling=C

The metabolic ledger W_met is an additive accounting function on the replication-certified class C_Bio^{nd} : $W_met(X \cup Y) = W_met(X) + W_met(Y)$ when X and Y are metabolically independent replication events. Additivity follows from the carrier-transport structure without importing thermodynamic primitives.

Dependencies. thm: BIO.REP

Discharge edges. ob: BIO.MET (full)

Proof stub. The metabolic ledger inherits additivity from the carrier-transport algebra established in thm: BIO.REP. The independence condition is carrier-algebraic: two replication events are metabolically independent when their carrier supports are disjoint in the replication record.

Theorem-like claim: Replication Fidelity Margin Theorem

Machine ID. thm: BIO.FID

Declared status. C **Computed status.** C

Scope. regime=nondegenerate-bio; domain=BIO-branch; certificate_scope=branch; promotion_ceiling=C

On the nondegenerate regime C_Bio^{nd} , there exists a positive fidelity margin $\delta_fid > 0$ separating the certified replicating class from the boundary of the non-replicating class. The margin is computable from the carrier parameters without invoking mutation-rate primitives.

Dependencies. thm: BIO.REP

Discharge edges. ob: BIO.FID (full)

Proof stub. Nondegeneracy of C_Bio^{nd} by definition provides the positive separation δ_fid . The separation is certified by the Book III depth-sum convergence applied to the replication-carrier chain: the depth sum for replicating objects converges to a limit strictly separated from the depth sum for non-replicating objects.

Theorem-like claim: No-Hidden Hereditary Channel Theorem

Machine ID. thm: BIO.NC

Declared status. C **Computed status.** C

Scope. regime=nondegenerate-bio; domain=BIO-branch; certificate_scope=branch; promotion_ceiling=C

Every hereditary pathway of a certified element of C_Bio^{nd} is visible in the declared replication kernel Φ_Bio . There is no undeclared hereditary channel: the carrier record is complete with respect to hereditary structure on the nondegenerate regime.

Dependencies. thm: BIO.REP

Discharge edges. ob: BIO.NC (full)

Proof stub. By the no-hidden-supplementation requirement of Book V branch-legitimacy (thm: VII.6.4), every structural pathway in a lawful branch must be declared in the branch's import ledger. Applied to hereditary pathways: any undeclared hereditary channel would constitute a prohibited import (Book VII, def: prohibited-import). Since Φ_Bio is the complete declared replication record, no channel outside Φ_Bio has canonical force in the branch.

Theorem-like claim: Hereditary Descent Transport Theorem

Machine ID. thm: BIO.HER

Declared status. C **Computed status.** C

Scope. regime=nondegenerate-bio; domain=BIO-branch; certificate_scope=branch; promotion_ceiling=C

The hereditary transport H_Bio passes the Book V branch-legitimacy test on C_Bio^{nd} : it descends from U through the declared replication functor with no hidden supplementation, and the descent structure is probe-response visible under the declared observation family.

Dependencies. thm: BIO.REP, thm: BIO.NC

Discharge edges. ob: BIO.HER (full)

Proof stub. Combine thm: BIO.REP (source-descent of the replication kernel) and thm: BIO.NC (no-hidden-channel). The hereditary transport is the carrier-push of the replication kernel through the declared descent map. Book V legitimacy follows: source-controlled (thm: BIO.REP), no hidden supplementation (thm: BIO.NC), probe-visible (nondegeneracy of C_Bio^{nd}).

Theorem-like claim: Boundary Self-Organization Theorem (Nondegenerate Regime)

Machine ID. thm: BIO.BND

Declared status. C **Computed status.** C

Scope. regime=nondegenerate-bio; domain=BIO-branch; certificate_scope=scoped-nondegenerate; promotion_ceiling=C; constraints=book:II-capacity-law

On the nondegenerate regime C_Bio^{nd} , boundary self-organization is sourced from the Book II capacity law: the interface structure B_Bio is the image of the bounded observational channel capacity bound applied to the replication-carrier boundary. No membrane biophysics, Lipschitz geometry, or cell-biology structural primitives are introduced.

Dependencies. thm: BIO.REP, thm: VII.6.4

Discharge edges. ob: BIO.BND (scoped)

Proof stub. Apply the Book II boundary capacity law (thm: VII.6.4 imports the Book II interface) to the replication carrier boundary: the capacity-limited interface produces a self-organized boundary structure in the image of the capacity bound. On C_Bio^{nd} the nondegeneracy condition guarantees that this image is non-trivial. The result is scoped to the nondegenerate regime: for the full replicator class the capacity bound may saturate at the boundary, leaving ob: BIO.BND-open as the residual frontier.

Theorem-like claim: BIO Endpoint on C_Bio^{nd}

Machine ID. thm: BIO.END

Declared status. C **Computed status.** C

Scope. regime=nondegenerate-bio; domain=BIO-branch; certificate_scope=branch; promotion_ceiling=C; constraints=ob: BIO.REP, ob: BIO.MET, ob: BIO.FID, ob: BIO.NC, ob: BIO.BND, ob: BIO.HER

The endpoint datum $E_Bio = (C_obs, O_Bio, F_Bio, B_Bio, W_met)$ is a certified lawful biology endpoint on the nondegenerate regime C_Bio^{nd} : certified replication kernel, additive metabolic ledger, positive fidelity margin, no-hidden-hereditary-channel control, boundary self-organization sourced through U, and hereditary descent transport satisfying Book V branch-legitimacy. The replication-founded-heredity endpoint is reached under these conditions. The endpoint is conditional on ob: BIO.REP through ob: BIO.HER; it does not claim evolutionary closure (ob: BIO.EVO) or full multicellular organization (ob: BIO.MULTI), which are downstream of the endpoint and may require Book VI bridge theorems.

Dependencies. thm: BIO.REP, thm: BIO.MET, thm: BIO.FID, thm: BIO.NC, thm: BIO.BND, thm: BIO.HER, thm: VII.6.4

Proof stub. Combine thm: BIO.REP through thm: BIO.HER. Apply the Book VII Shared Endgame Schema (prop: shared-endgame via thm: VII.6.4): part (i) is the source-controlled realized object given by the replication kernel descended from U; part (ii) is the metabolic ledger theorem together with the hereditary descent theorem; part (iii) is the no-hidden-channel theorem together with the fidelity margin. Full proof is conditional on ob: BIO.REP through ob: BIO.HER. The result is scoped to C_Bio^{nd} : evolutionary closure (ob: BIO.EVO) and multicellular organization (ob: BIO.MULTI) are downstream and not claimed here.

Theorem-like claim: Minimal Two-Unit Assembly Theorem

Machine ID. thm:BI0.MULTI.twoocell

Declared status. C **Computed status.** C

Scope. regime=nondegenerate-bio; domain=BIO-branch; certificate_scope=branch; promotion_ceiling=C; constraints=two-cell-calibration-analogue

The minimal faithful coordination carrier $K^{\text{coord}}_{\{A\textsubscript{2}\}}$ for the two-unit assembly $A\textsubscript{2} = \{X\textsubscript{1}, X\textsubscript{2}\}$ consists of exactly the metabolic exchange carrier m_{coord} and the hereditary channel carrier h_{coord} . Both are load-bearing: removing either destroys the inter-unit coordination linkage. The coordination ledger is additive over the two-unit assembly. This result is the BIO analogue of the caesium calibration nomination in SPEC: it declares the minimal assembly structure rather than solving full multicellular organization.

Dependencies. thm:BI0.END

Proof stub. The two-unit assembly is the minimal coordination carrier satisfying: (1) both units are in $C_{\text{Bio}}^{\text{nd}}$; (2) the inter-unit metabolic exchange m_{coord} touches both replication-maintaining cores (from thm:BI0.MET applied to each unit); (3) the inter-unit hereditary channel h_{coord} is declared and no-hidden (from thm:BI0.NC). Minimality: removing m_{coord} destroys inter-unit metabolic accounting; removing h_{coord} destroys inter-unit hereditary linkage. Additivity of the coordination ledger follows from thm:BI0.MET applied to the joint system.

Frontier: Boundary Self-Organization (Full Replicator Class)

Machine ID. ob:BI0.BND-open

Declared status. O **Computed status.** O

Scope. regime=full-replicator-class; domain=BIO-branch; certificate_scope=open-class; promotion_ceiling=C

The boundary obligation for the full replicator class — beyond the nondegenerate regime $C_{\text{Bio}}^{\text{nd}}$ — remains open. Whether every replicator (including those at or near the nondegeneracy boundary) admits source-descended boundary self-organization is a genuine force question whose resolution may require either a new ingredient or a Transfer Theorem from Book VI.

Status note: BIO Branch Posture

Machine ID. status:BI0

Declared status. conditional-cert-close **Computed status.** conditional-cert-close

Scope. regime=nondegenerate-bio; domain=BIO-branch; certificate_scope=branch

BIO branch posture: conditional CERT-CLOSE on the nondegenerate certified biological class $C_{\text{Bio}}^{\text{nd}}$, with two-unit assembly declared as the minimal coordination structure.

The branch has discharged obligations REP, MET, FID, NC, HER conditionally, and established the scoped discharge of BND on the nondegenerate regime.

Active frontier: ob:BI0.BND-open (BND for the full replicator class) remains genuinely open. This is not a Clay-grade arithmetic item but a structural force question about whether the full class admits source-descended boundary self-organization without a Book VI bridge.

The no-smuggling boundary is strictly maintained: the branch derives all structure from role carriers and source descendants. No genes, cells, fitness functions, thermodynamic primitives, or Lipschitz membrane geometry are imported as assumptions.

Dependencies. thm:BI0.END, thm:V.UBLT, thm:VII.6.4